AI Used for Deliverable 2

Kiki (Task 1 and 3)

1. File reading (`Path.read_bytes`)

`f.read_bytes()` reads a whole file into memory as `bytes` in one open/close, so you can decode or hash it without reopening the file.

2. Finding the data files

Your `Path("d2_data\\2026").rglob("*")` used a Windows-only separator and a path relative to the working directory. We switched to building paths with `/` from a base folder. `BASE_DIR = ...` only builds a path; it doesn't create the folder (that needs `mkdir`).

3. The `__file__` NameError

You're running in Jupyter, which doesn't define `__file__`. The fix is `Path.cwd()` instead, plus a check that raises a clear error if no files are found.

4. The log parser

`parse_log()` using a verbose regex with named groups to capture timestamp, register_id, item, price, currency, and status, ignoring noise lines.

5. Frequency counting with dictionaries

Three approaches: manual `.get(key, 0) + 1`, `defaultdict(int)`, and `Counter` (best, with `.most_common()`).

6. `compute_refund_ratio`

The function was fine (`refund_count / (sales_count + EPS)`). The bug was in your code: you built `register_ids` from the `item` column instead of `register_id`, which made every count zero. I also advised against overwriting your list with a DataFrame of itself.

Thea (Task 2 and 4)

1. Used for explaining virtual environment and how it relates to the requirements.txt file.
2. Write test functions for task 4 and generate test data
3. Explain pandas functions such as `resample` and `groupby`
4. How to access the right file path